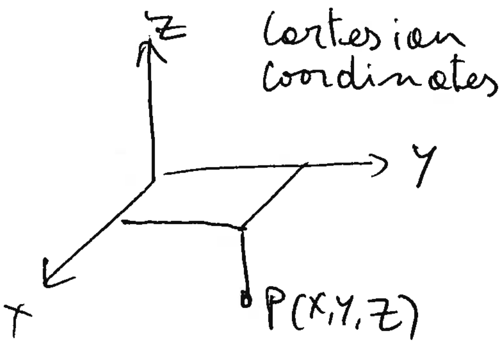
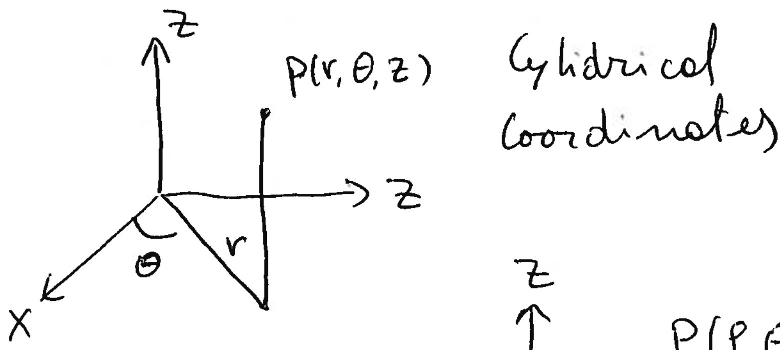
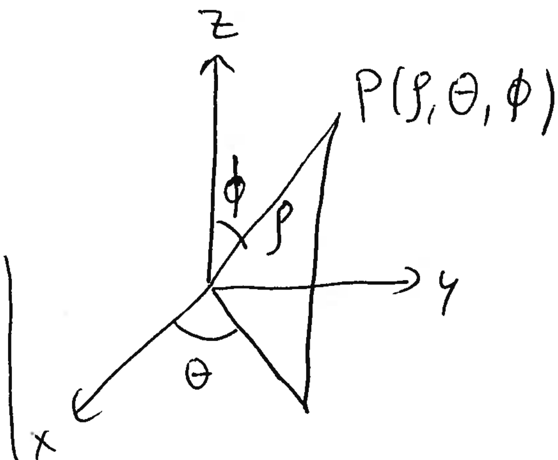
§11.9 Cylindrical & Spherical Coordinates





$0 \le \theta < 2\pi$



$\rho \geq 0 \quad 0 \le \theta < 2\pi$

$0 \le \phi \le \pi$

Cylindrical to Cartesian

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$$\left.\begin{array}{l} r = 2 \\ \theta = \pi/3 \\ z = 4 \end{array}\right\} \Rightarrow \begin{array}{l} x = 2\cos\pi/3 = 1 \\ y = 2\sin\pi/3 = \sqrt{3} \\ z = 4 \end{array}$$

Cartesian to Cylindrical

$r = \sqrt{x^2+y^2} \qquad \tan\theta = y/x \qquad z = z$

$$\left.\begin{array}{l} x = 2 \\ y = 3 \\ z = 4 \end{array}\right\} \Rightarrow \begin{array}{l} r = \sqrt{4+9} = \sqrt{13} \\ \theta = \tan^{-1}(3/2) \\ z = 4 \end{array}$$

Ex eq of a sphere $x^2+y^2+z^2 = 25$

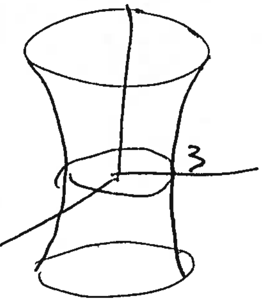
$\Rightarrow r^2\cos^2\theta + r^2\sin^2\theta + z^2 = 25 \quad \Rightarrow \quad r^2+z^2 = 25$

paraboloid $x^2+y^2 = 4-z \quad \Rightarrow r^2 = 4-z$

cylinder $x^2+y^2+2y = 0 \quad \Rightarrow r^2 + 2r\sin\theta = 0$

$\Rightarrow r = -2\sin\theta$

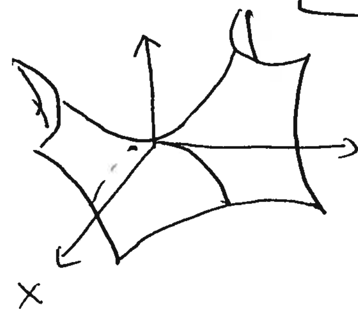
Ex $r^2 - z^2 = 9 \implies x^2 + y^2 - z^2 = 9$



Ex $r^2 \cos 2\theta = z \implies r^2(\cos^2\theta - \sin^2\theta) = z$

$\implies x^2 - y^2 = z$

Hyperbolic Paraboloid



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spherical to Cartesian

$x = \rho \sin\phi \cos\theta$

$y = \rho \sin\phi \sin\theta$

$z = \rho \cos\phi$

Cartesian to Spherical

$\rho = \sqrt{x^2 + y^2 + z^2}$

$\tan\theta = y/x$

$\cos\phi = \dfrac{z}{\sqrt{x^2+y^2+z^2}}$

$\begin{array}{c} \rho \quad \theta \quad \phi \\ (8, \pi/3, \pi/4) \end{array} \rightarrow$

$x = 8 \sin \pi/4 \cos \pi/3 = 8 \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = 2\sqrt{2}$

$y = 8 \sin \pi/4 \sin \pi/3 = 8 \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} = 2\sqrt{6} \rightarrow (2\sqrt{2}, 2\sqrt{6}, 4\sqrt{2})$

$z = 8 \cos \pi/4 = 8 \frac{\sqrt{2}}{2} = 4\sqrt{2}$

$\begin{array}{c} x \quad y \quad z \\ (1, 2, 3) \end{array} \rightarrow$

$\rho = \sqrt{1+4+9} = \sqrt{14}$

$\theta = \tan^{-1} 2$ ... use calculator ....

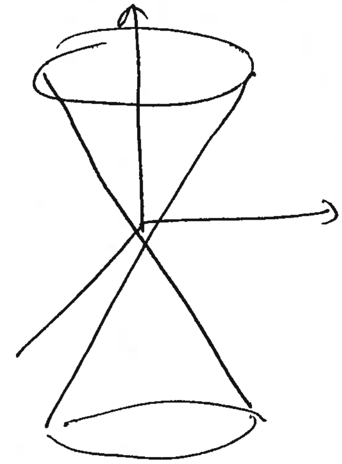
$\phi = \cos^{-1}\left(\frac{3}{\sqrt{14}}\right)$

Ex $\rho = \cos\phi \implies \sqrt{x^2+y^2+z^2} = \dfrac{z}{\sqrt{x^2+y^2+z^2}} \implies x^2+y^2+z^2 = z$

$\implies x^2 + y^2 + (z-1)^2 = 1$ Sphere of radius 1 & center at $(0,0,1)$

Ex $\quad x^2+y^2=z^2 \Rightarrow x^2+y^2+z^2=2z^2$

$\Rightarrow \rho^2 = 2\rho^2\cos\phi \Rightarrow \cos\phi = 1/2$



Ex $\quad z = x^2+y^2 \rightarrow \rho\cos\phi = \rho^2\sin^2\phi\cos^2\theta + \rho^2\sin^2\phi\sin^2\theta$

$$= \rho^2\sin^2\phi$$

$$\Rightarrow \cos\phi = \rho\sin^2\phi \Rightarrow \rho = \frac{\cos\phi}{\sin^2\phi}$$

Exercises

① Change $(1, \pi/2, 1)$ & $(-2, \pi/4, 2)$ from cylindrical to spherical coords.

② Change $(8, \pi/4, \pi/6)$ from spherical to ~~cylinder~~ Cartesian coords.

③ Sketch the following:

- $r=5$ • $\rho=5$ • $\phi=\pi/6$ • $\theta=\pi/6$
- $r=3\cos\theta$ • $r=2\sin 2\theta$ • $r^2+z^2=9$

④ Change $x^2+y^2=9$ to cylindrical an spherical coords

⑤ Repeat ④ for $x^2-y^2=25$ & $x^2+y^2+4z^2=10$

⑥ Change $r^2\cos 2\theta = z$ & $\rho\sin\phi=1$ to Cartesian coords.